Modern Academy
for Engineering and Technology
Basic Science Department

Academic Year: 2020-2021
Semester: Fall
Exam Date: 6/12/2020

QUESTIONS FOR MID-TERM EXAM

Subject: (MTH 203) Differential Equations and Transforms **Spec:** Comm. and Comp.
Examiners: Assoc. Prof. Ashraf Taha & Prof. Aly Essawi **Time:** 60 minutes

Question 1: Find the general solution of the first ordinary differential equations

a) $y' = \sec y + \dfrac{x}{\cos y}$.

b) $y' = (x+y)\ln(x+y) - 1$.

c) $(\sin x \cos y + xe^x)dx + (y^2 + \cos x \sin y)dy = 0$.

d) $y' + y = e^{2x} \tan x \ \ y^3$.

Question 2: Find the general solution of the following differential equations

a) $y''' - 5y'' + 7y' - 3y = 0$.

b) $y'' - 2y' - 3y = 2xe^x$.

c) $\dfrac{1}{x}y'' + \dfrac{6}{x^2}y' - \dfrac{6y}{x^3} = 12x^3 + 9$, given that $y = x$ is a solution of $\dfrac{1}{x}y'' + \dfrac{6}{x^2}y' - \dfrac{6y}{x^3} = 0$.

GOOD LUCK

Modern Academy
for Engineering and Technology
Basic Science Department

Academic Year: 2020-2021
Semester: Fall
Exam Date: 6/12/2020

QUESTIONS FOR MID-TERM EXAM

Subject: (MTH 203) Differential Equations and Transforms **Spec:** Comm. and Comp.
Examiners: Assoc. Prof. Ashraf Taha & Prof. Aly Essawi **Time:** 60 minutes

Question 1: Find the general solution of the first ordinary differential equations

a) $y' = \sec y + \dfrac{x}{\cos y}$.

b) $y' = (x+y)\ln(x+y) - 1$.

c) $(\sin x \cos y + xe^x)dx + (y^2 + \cos x \sin y)dy = 0$.

d) $y' + y = e^{2x} \tan x \ \ y^3$.

Question 2: Find the general solution of the following differential equations

a) $y''' - 5y'' + 7y' - 3y = 0$.

b) $y'' - 2y' - 3y = 2xe^x$.

c) $\dfrac{1}{x}y'' + \dfrac{6}{x^2}y' - \dfrac{6y}{x^3} = 12x^3 + 9$, given that $y = x$ is a solution of $\dfrac{1}{x}y'' + \dfrac{6}{x^2}y' - \dfrac{6y}{x^3} = 0$.

GOOD LUCK